Life Satisfaction Analysis

Your Name

September 23, 2025

1 Introduction

This document presents tables and figures for life satisfaction analysis across domains (work, health, income, leisure), including full sample and subgroup results (gender, East/West, migration background).

2 Data

For this analysis, we rely on long-run data on domain satisfaction from the Socio-Economic Panel (SOEP). The SOEP is a representative panel of German households which started in 1984 and follows households as well as their members on a yearly basis. The SOEP surveys it's respondents on a wide range of topics, such as their labor market experience, their household situation and attitudes.

Important for our analysis, the SOEP contains long running individual level time series of domain satisfactions. On a yearly basis, individuals are asked about their satisfaction with a wide range of life domains. For our analysis, we focus on a subset of these. Specifically, we focus on satisfaction with work, income, health, and leisure. The reason for this is that these dimension characterize the labor supply decisions of individuals. At the same time, we believe that these dimensions very broadly characterize large life dimensions, which subsume a lot of other domains, such as sleep, time with children, just to name two examples. Specifically, individuals are asked "How satisfied are you today with the following areas of your life? Please answer on a scale from 0 to 10, where 0 means completely dissatisfied and 10 means completely satisfied." The domain prompts then read

- "… with your health?"
- "… (if employed) with your job?"
- "… with your household income?"
- "… with the amount of leisure time/your leisure time?"

Further content:

- restrictions
- Relevance in the lit
- baseline categories

3 Empirical Strategy

Our empirical strategy follows the age–period–cohort (APC) decomposition pioneered by DeatonPaxson1994 to recover life-cycle patterns of domain satisfactions from the SOEP data, which we treat as repeated cross-sections. We construct *synthetic cohorts* defined by year of birth. For each survey year, we compute cohort means of the outcomes of interest (e.g., satisfaction with work, income, leisure, and health) by age. Tracking these cohort means across survey waves yields a pseudo-panel that can be analyzed in a life-cycle framework.

Formally, let $y_{a,b,t}$ denote the cohort-average outcome for individuals of age a, born in cohort b, observed in calendar year t. We model outcomes as the sum of an age effect α_a, a cohort effect γ_b, and a period effect δ_t:

$$y_{a,b,t} = \alpha_a + \gamma_b + \delta_t + \varepsilon_{a,b,t}. \tag{1}$$

Age effects capture the life-cycle evolution of domain satisfactions, cohort effects capture differences in lifetime resources and demographic environments across birth cohorts, and period effects capture macroeconomic shocks that affect all individuals simultaneously.

A well-known difficulty is that age, period, and cohort are linearly dependent ($b = t - a$), so their effects cannot be separately identified without additional restrictions. We resolve this identification problem by imposing the normalization used by DeatonPaxson1994: we require that period effects sum to zero and are orthogonal to a linear time trend. Intuitively, this restriction attributes long-run secular changes to cohorts, while allowing period effects to capture only transitory fluctuations. We further impose that each set of effects sums to zero, which makes them interpretable as deviations from the grand mean.

Empirically, we estimate Equation (1) using ordinary least squares on cohort-level averages of log outcomes. The regression includes a full set of age, cohort, and period indicators, subject to the constraints described above. Each cell is weighted by the square root of the number of individuals used to construct the cohort mean, which accounts for heteroskedasticity in the measurement error of cohort averages. Implementation is carried out in Stata using the `apc.ado` routine, which automates the construction of age, period, and cohort dummies, imposes the identification restrictions, and produces estimates of the three profiles.

This procedure yields three distinct sets of estimates: (i) the age profile, which traces life-cycle variation in domain satisfactions; (ii) the cohort profile, which reflects differences in economic resources and demographic conditions across generations; and (iii) the period profile, which captures short-run shocks common to all cohorts. The decomposition allows us to distinguish between life-cycle changes, long-run cohort differences driven by growth and demographic transition, and transitory shocks.

4 Descriptive Table of Satisfaction Variables

Table 1: Descriptive Statistics for Satisfaction Variables

	(1)		
	mean	sd	count
Lebenszufriedenheit gegenwaertig	7.16	1.81	813603.00
Zufriedenheit Gesundheit	6.78	2.26	810866.00
Zufriedenheit HH-Einkommen	6.53	2.29	766740.00
Zufriedenheit Wohnung	7.72	2.03	772355.00
Zufriedenheit Freizeit	7.01	2.24	721121.00
Zufriedenheit Arbeit	7.10	2.11	480387.00
Zufriedenheit HH-Taetigk.	6.78	2.01	577087.00
Zufriedenheit mit persoenlichem Einkommen	6.13	2.56	492085.00
Zufriedenheit Kinderbetreuung	6.96	2.50	131374.00
Zufriedenheit Schlaf	6.76	2.25	392401.00
Zufriedenheit Familienleben	7.92	1.94	438528.00
Zufriedenheit: Netz sozialer Sicherung in der BRD	6.10	2.21	152013.00
Zufriedenh. Schul- und Berufsausbildung	7.25	2.13	130791.00
Zufriedenheit Freizeitumfang	6.64	2.46	23156.00
Zufriedenheit Lebensstandard	7.13	1.91	341423.00
Zufriedenh. Wohngegend	7.56	2.13	72074.00
Zufriedenheit: Waren- / Dienstleistungsangebot	6.44	2.62	198531.00
Zufriedenh. Umweltzustand	6.24	2.13	219855.00
Zufriedenh. OePNV	6.36	2.56	36831.00
Zufriedenheit mit Demokratie [2005, 2010, 2016, 2020, 2021]	5.99	2.45	114870.00
Zufriedenheit Freundes-, Bekanntenkreis	7.72	1.85	86619.00
Zufriedenheit ehrenamtl. Taetigkeit	6.82	2.50	9792.00
Zufriedenheit mit Verwirklichung sozialer Gerechtigkeit in D	5.05	2.23	24881.00
Average domain satisfaction	6.94	1.40	816269.00
Observations	816269		

5 Full Sample Figures

Figure 1: Satisfaction with work (full sample)

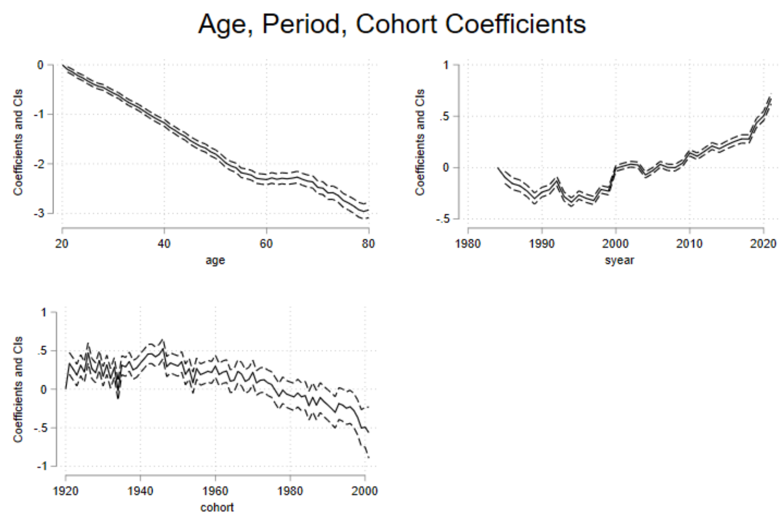


Figure 2: Satisfaction with health (full sample)

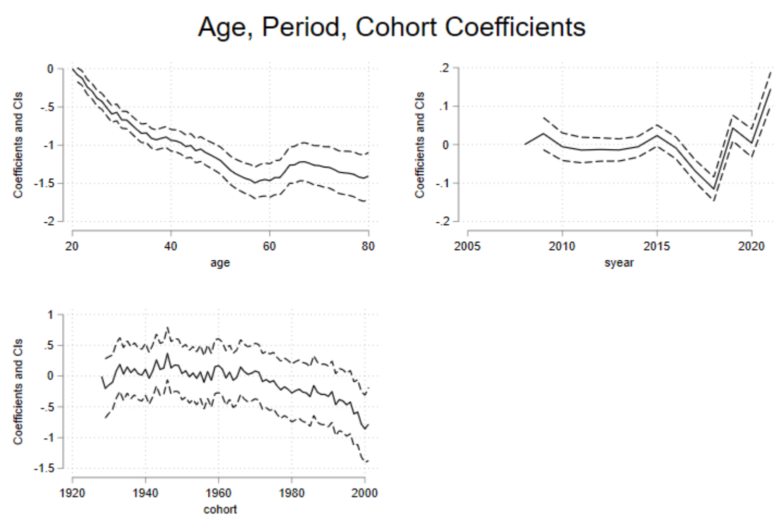


Figure 3: Satisfaction with income (full sample)

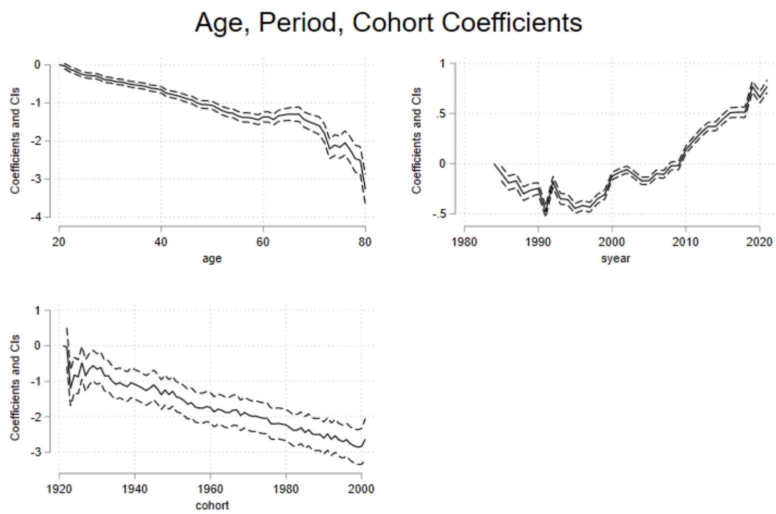
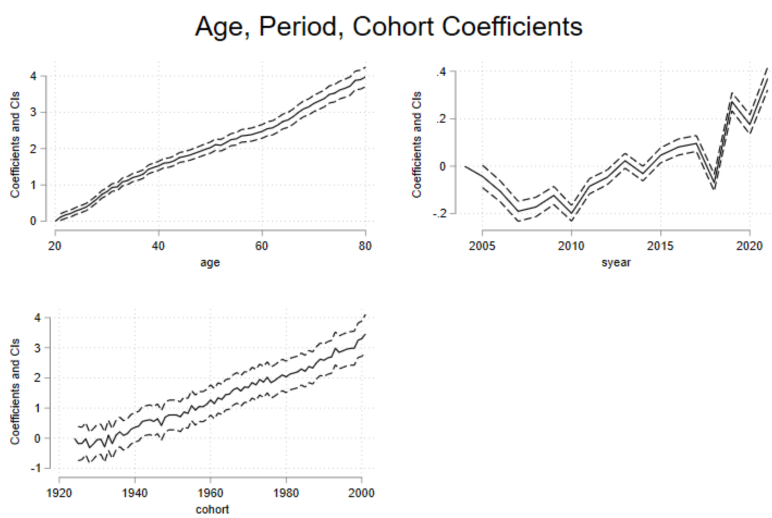


Figure 4: Satisfaction with leisure (full sample)



6 APC Overlay Figures by Subgroup

This section presents separate APC profiles (age, period, cohort) for each satisfaction domain and subgroup comparison.

Figure 5: Satisfaction with Work (Gender — Age Profile)

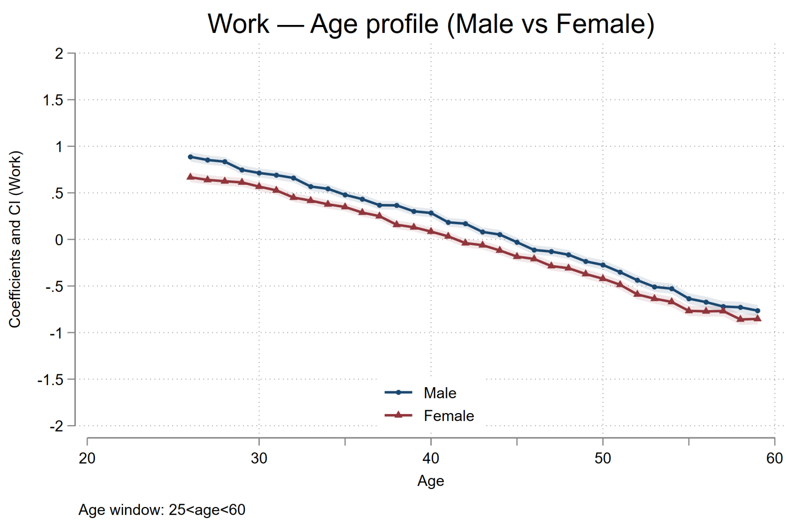


Figure 6: Satisfaction with Work (Gender — Period Profile)

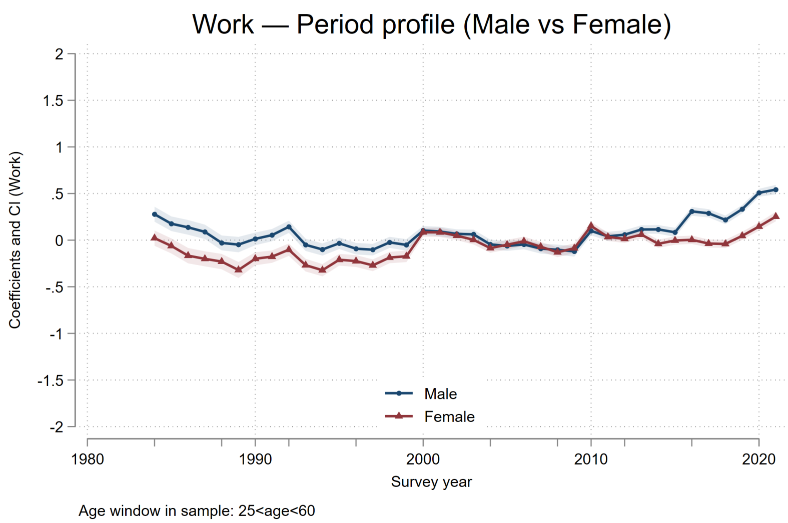


Figure 7: Satisfaction with Work (Gender — Cohort Profile)

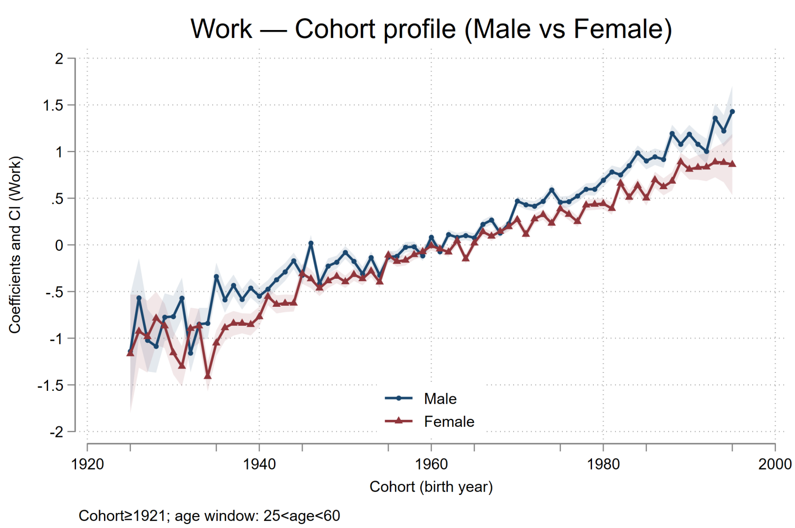


Figure 8: Satisfaction with Work (East vs West — Age Profile)

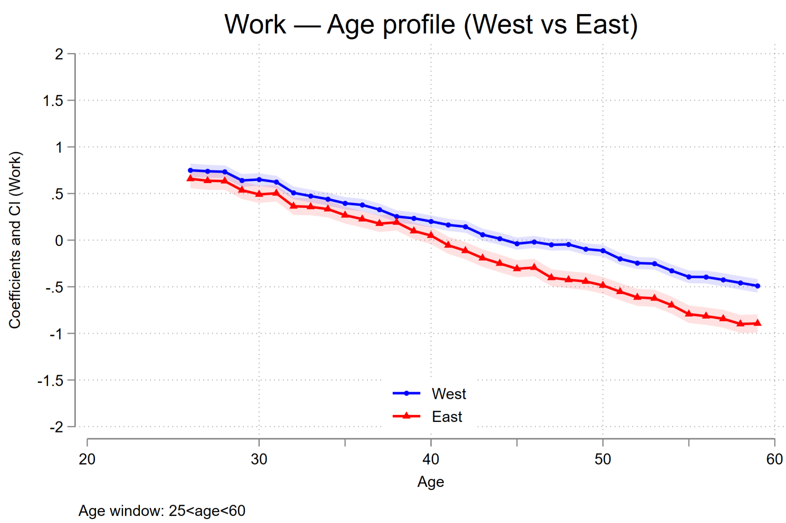


Figure 9: Satisfaction with Work (East vs West — Period Profile)

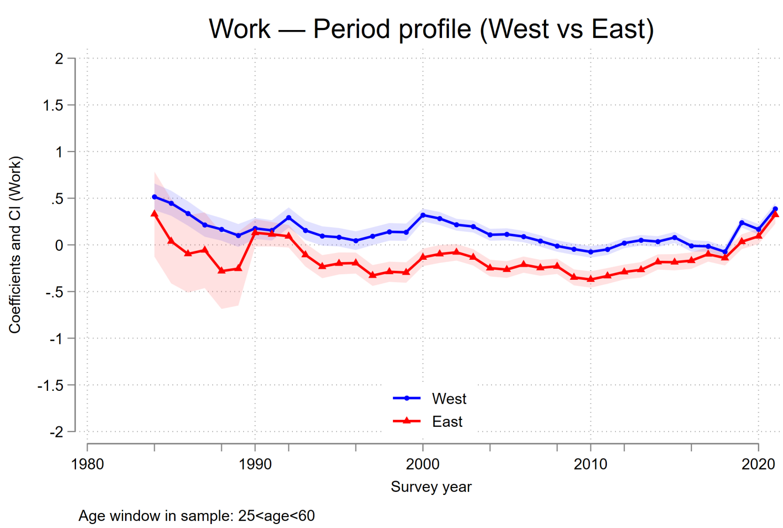


Figure 10: Satisfaction with Work (East vs West — Cohort Profile)

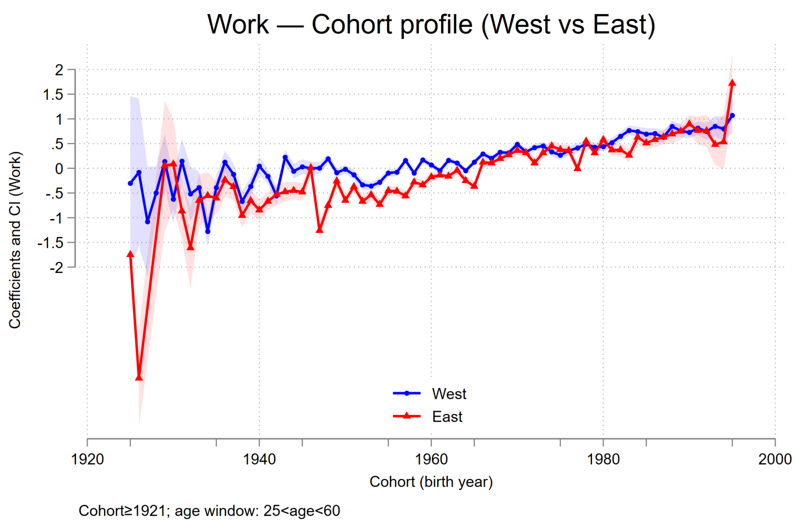


Figure 11: Satisfaction with Work (Migration Background — Age Profile)

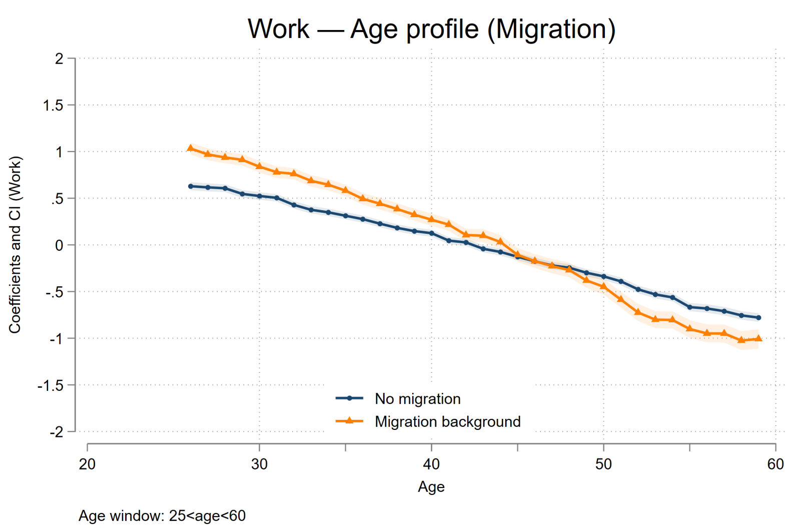


Figure 12: Satisfaction with Work (Migration Background — Period Profile)

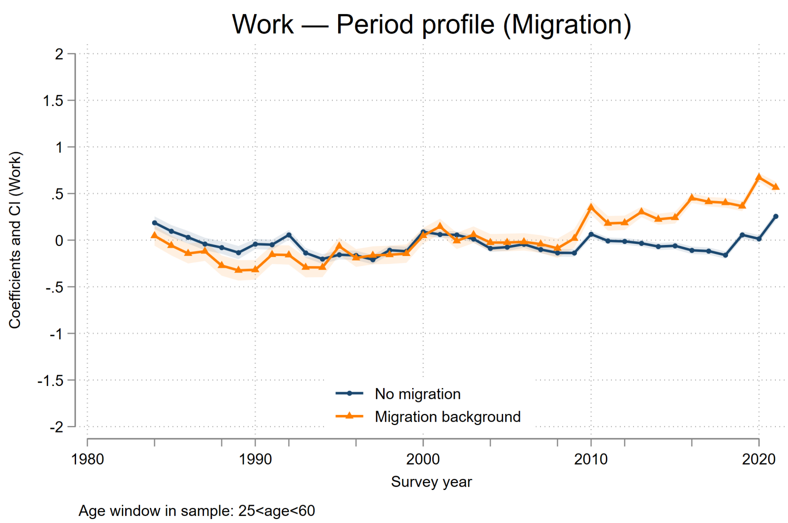


Figure 13: Satisfaction with Work (Migration Background — Cohort Profile)

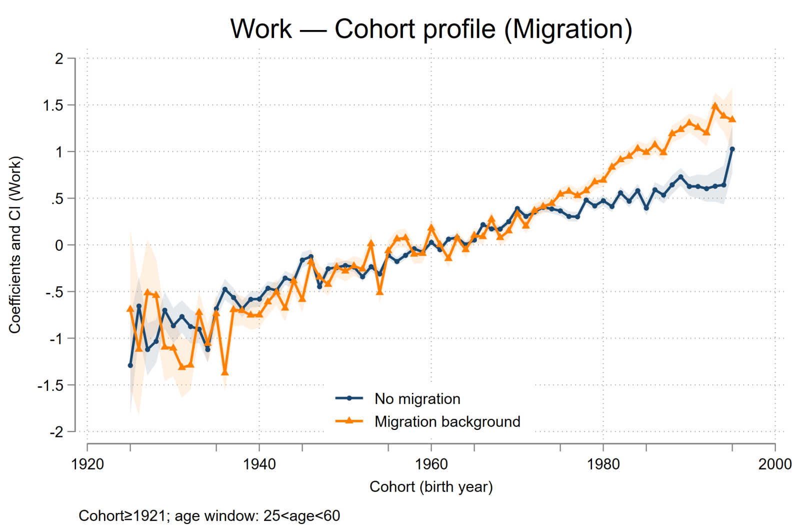


Figure 14: Satisfaction with Income (Gender — Age Profile)

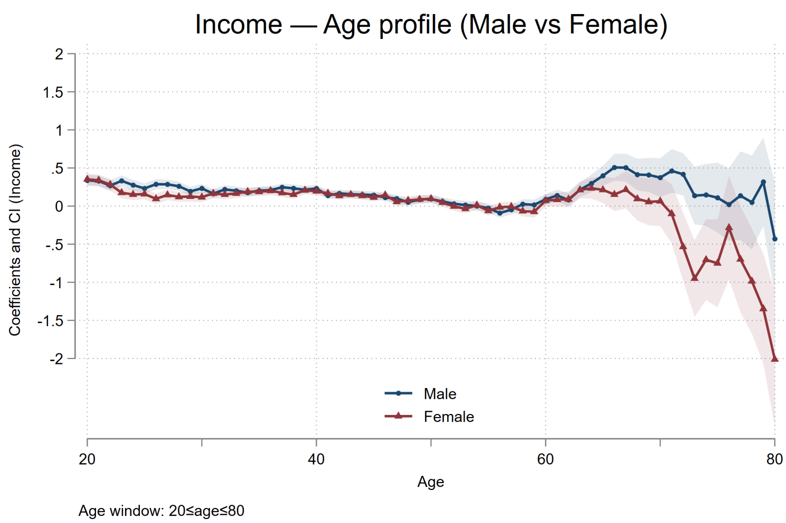


Figure 15: Satisfaction with Income (Gender — Period Profile)

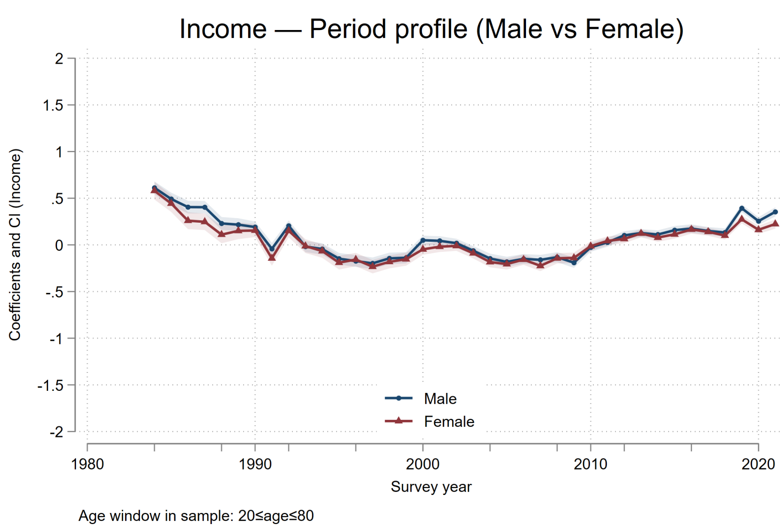


Figure 16: Satisfaction with Income (Gender — Cohort Profile)

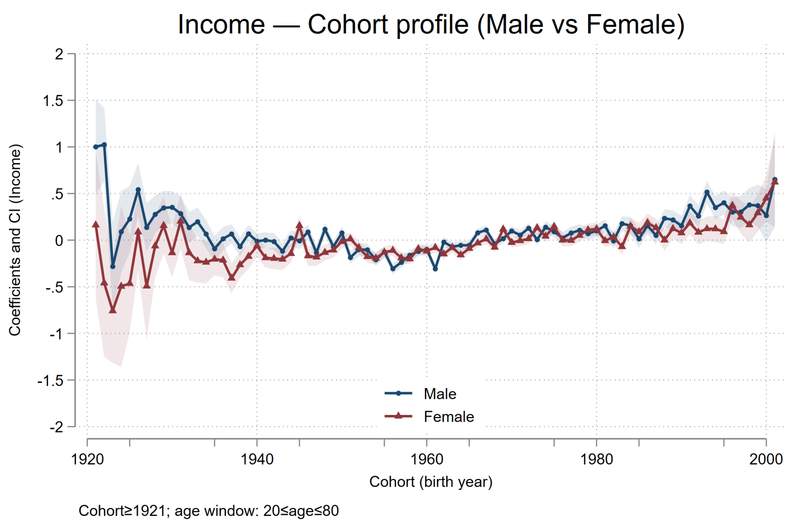


Figure 17: Satisfaction with Income (East vs West — Age Profile)

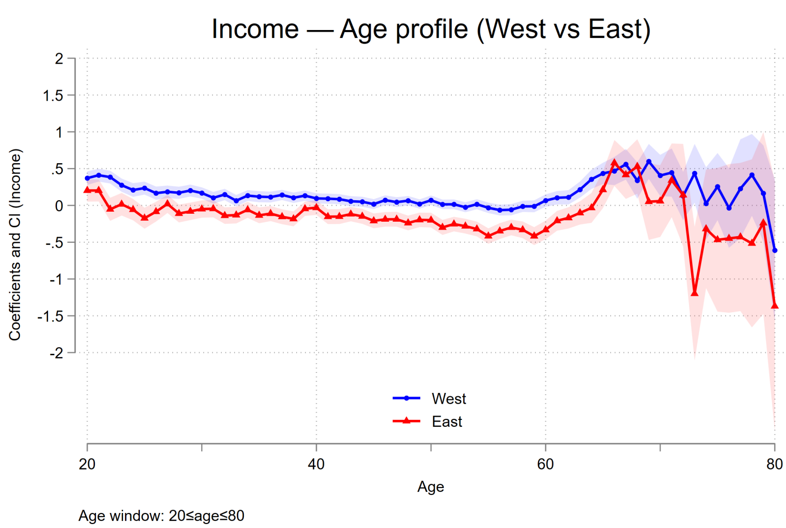


Figure 18: Satisfaction with Income (East vs West — Period Profile)

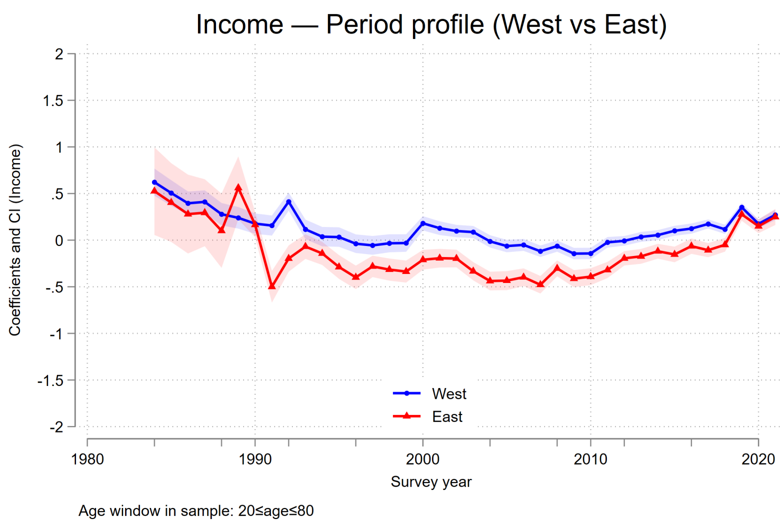


Figure 19: Satisfaction with Income (East vs West — Cohort Profile)

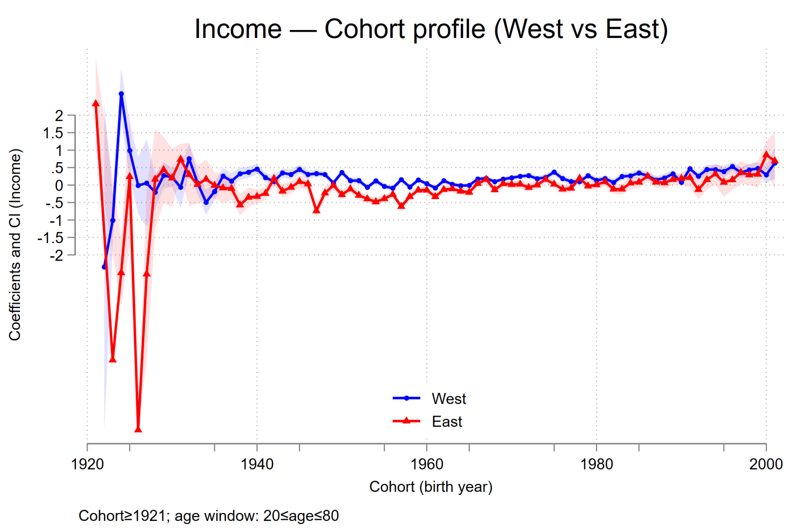


Figure 20: Satisfaction with Income (Migration Background — Age Profile)

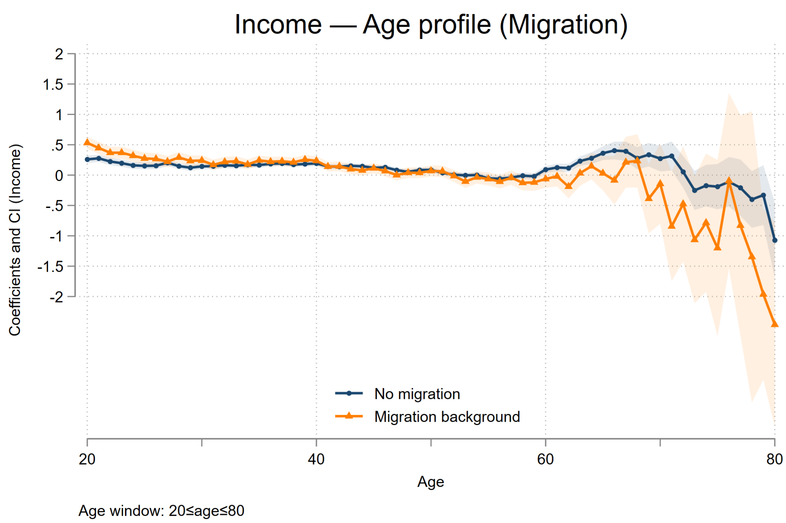


Figure 21: Satisfaction with Income (Migration Background — Period Profile)

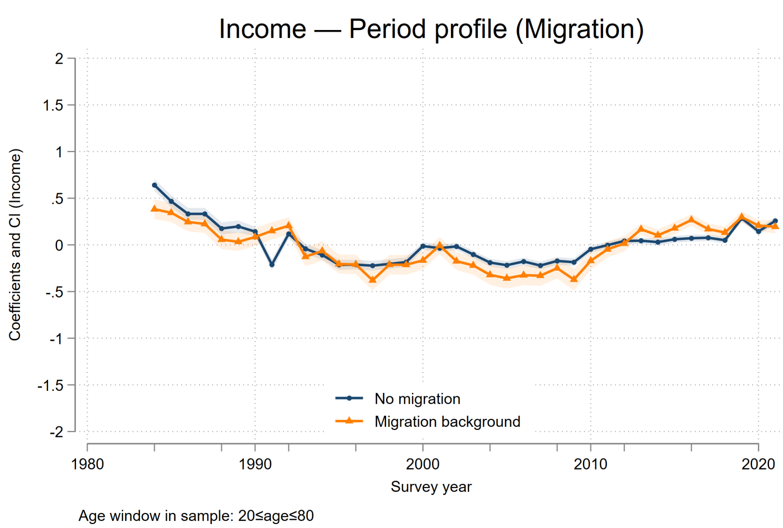


Figure 22: Satisfaction with Income (Migration Background — Cohort Profile)

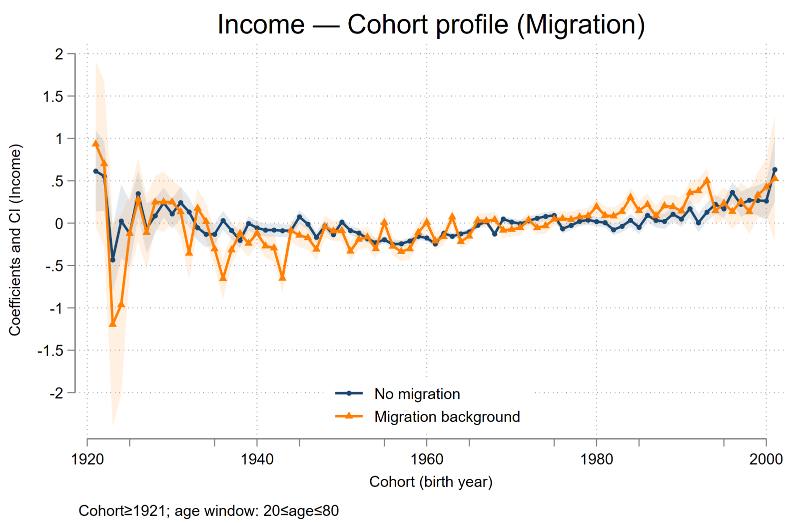


Figure 23: Satisfaction with Health (Gender — Age Profile)

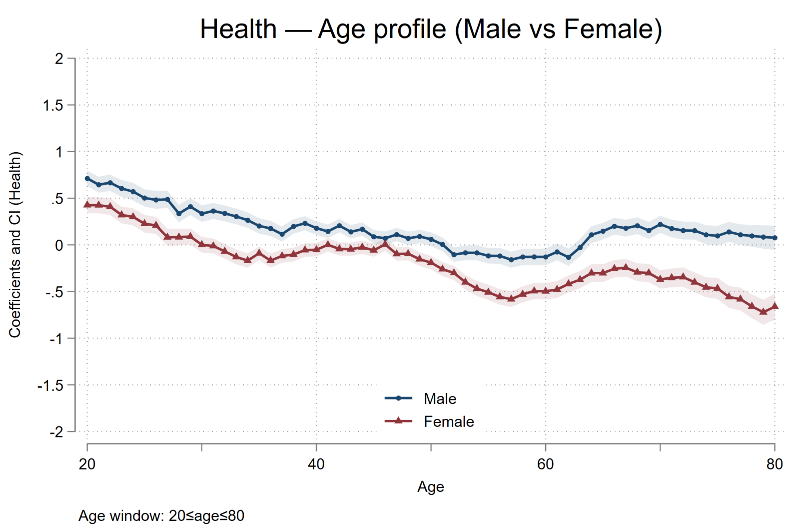


Figure 24: Satisfaction with Health (Gender — Period Profile)

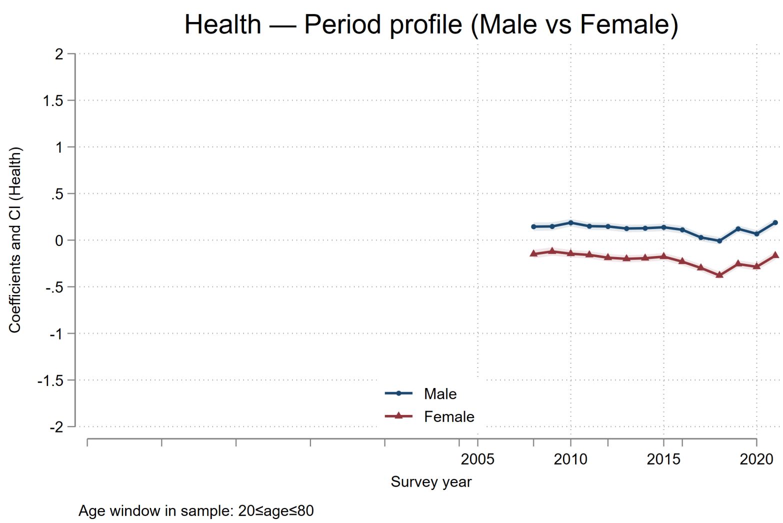


Figure 25: Satisfaction with Health (Gender — Cohort Profile)

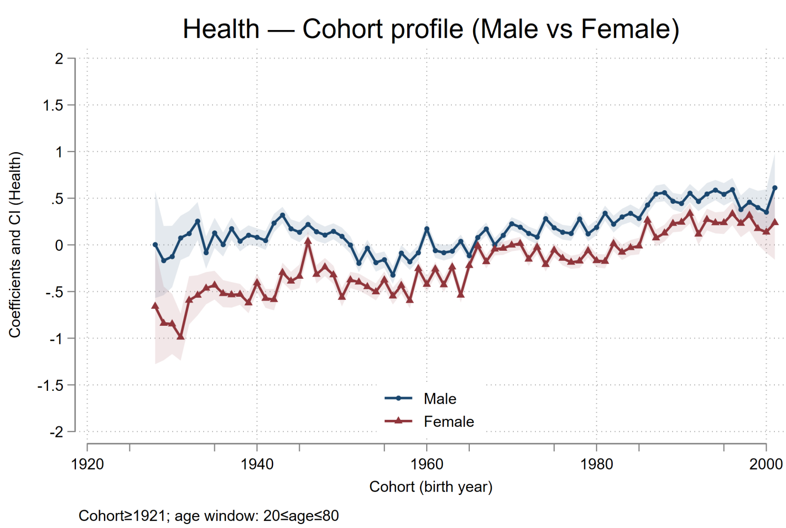


Figure 26: Satisfaction with Health (East vs West — Age Profile)

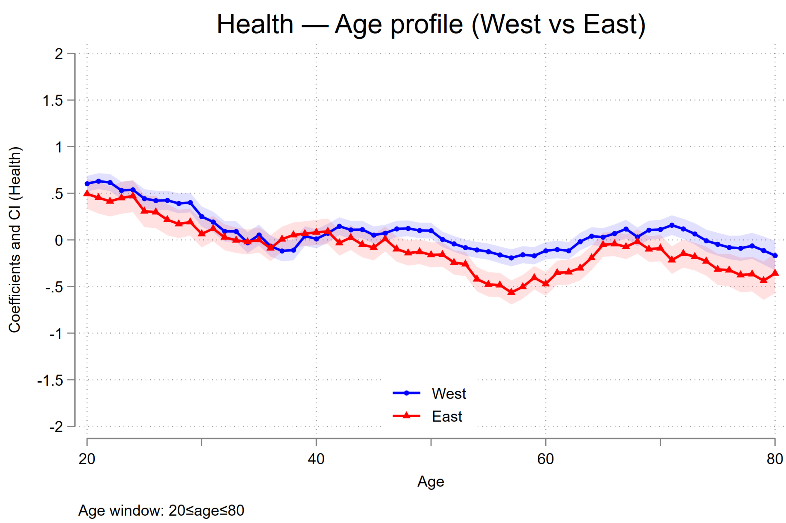


Figure 27: Satisfaction with Health (East vs West — Period Profile)

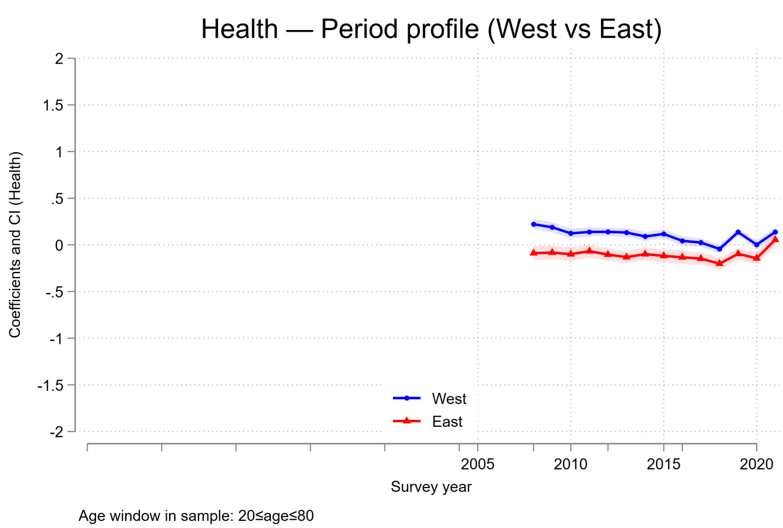


Figure 28: Satisfaction with Health (East vs West — Cohort Profile)

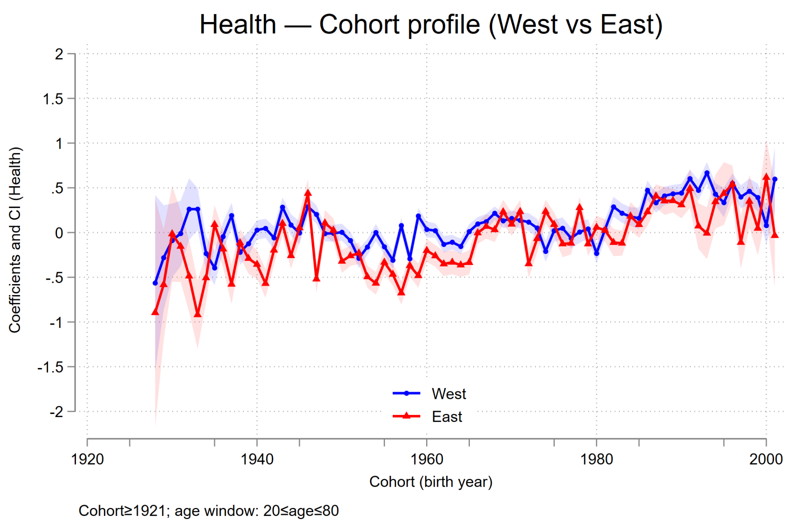


Figure 29: Satisfaction with Health (Migration Background — Age Profile)

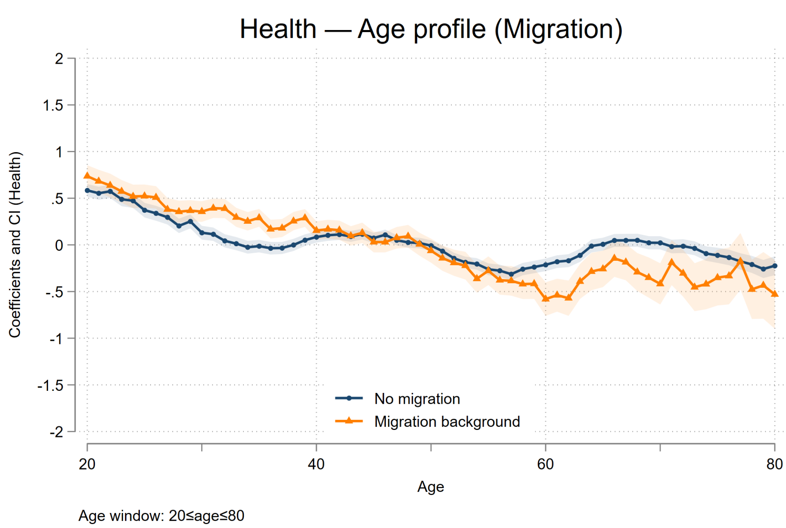


Figure 30: Satisfaction with Health (Migration Background — Period Profile)

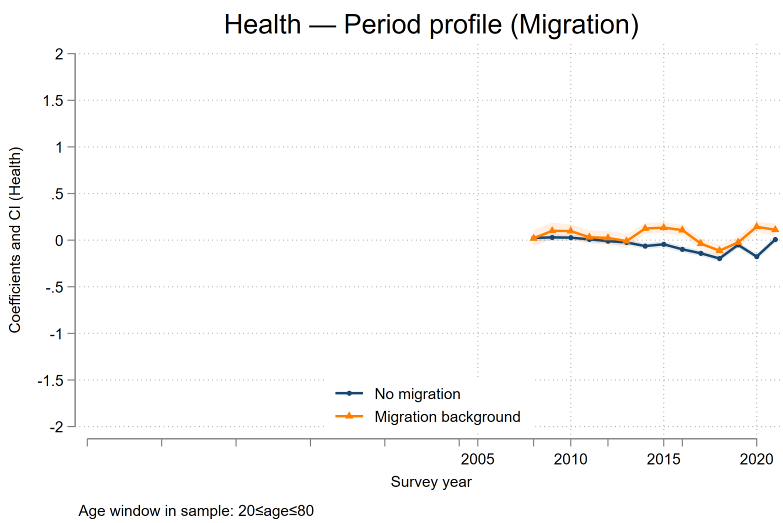


Figure 31: Satisfaction with Health (Migration Background — Cohort Profile)

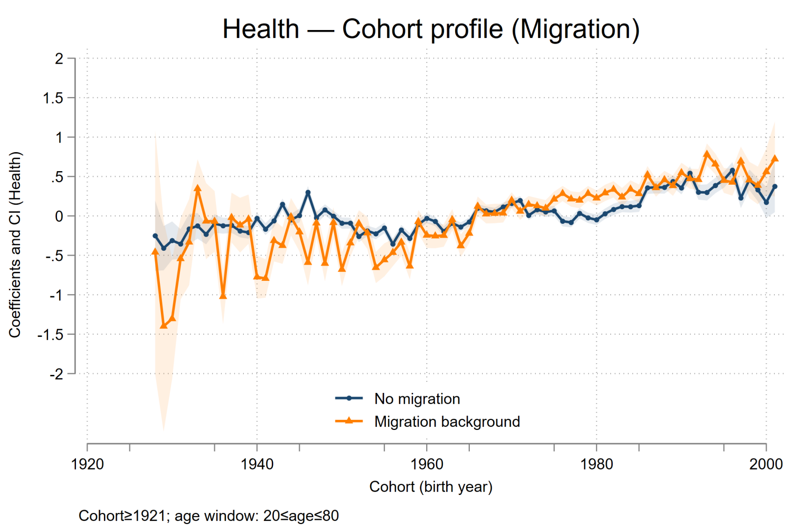


Figure 32: Satisfaction with Leisure (Gender — Age Profile)

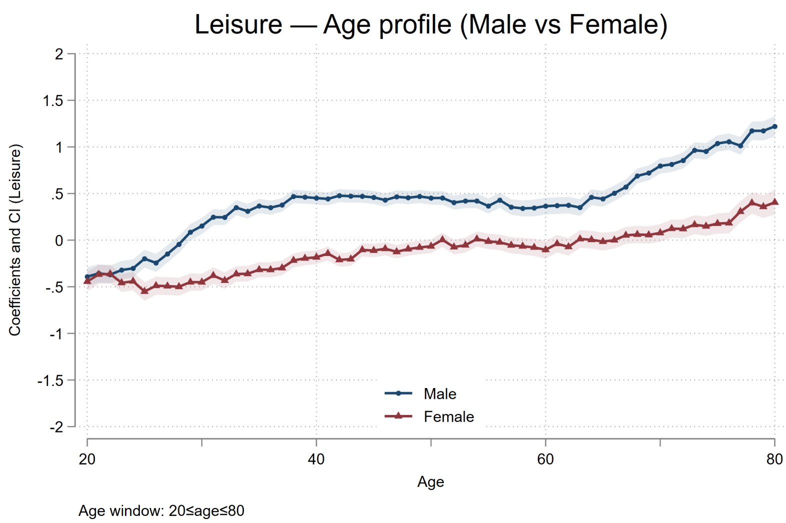


Figure 33: Satisfaction with Leisure (Gender — Period Profile)

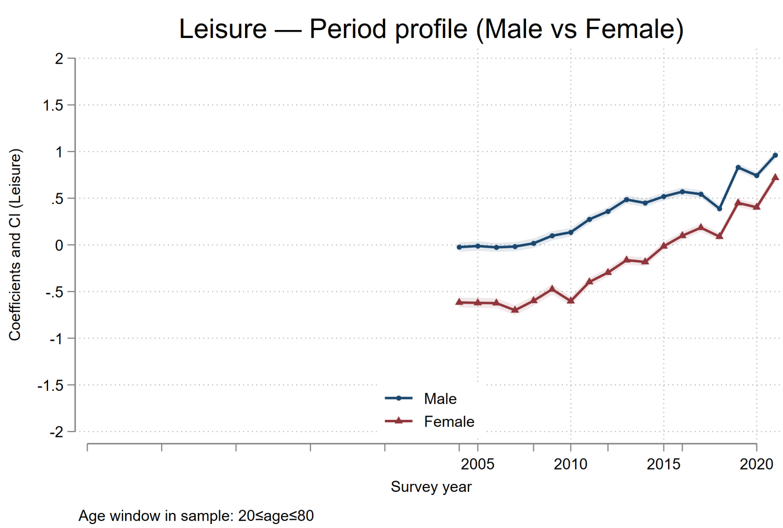


Figure 34: Satisfaction with Leisure (Gender — Cohort Profile)

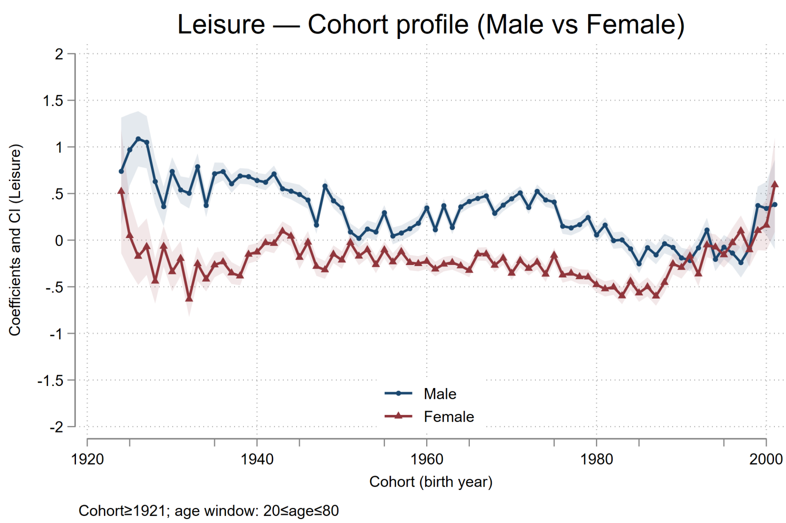


Figure 35: Satisfaction with Leisure (East vs West — Age Profile)

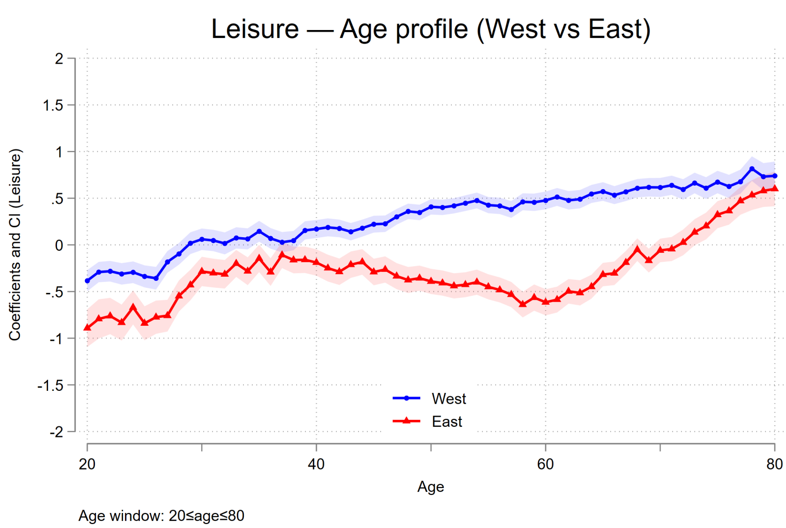


Figure 36: Satisfaction with Leisure (East vs West — Period Profile)

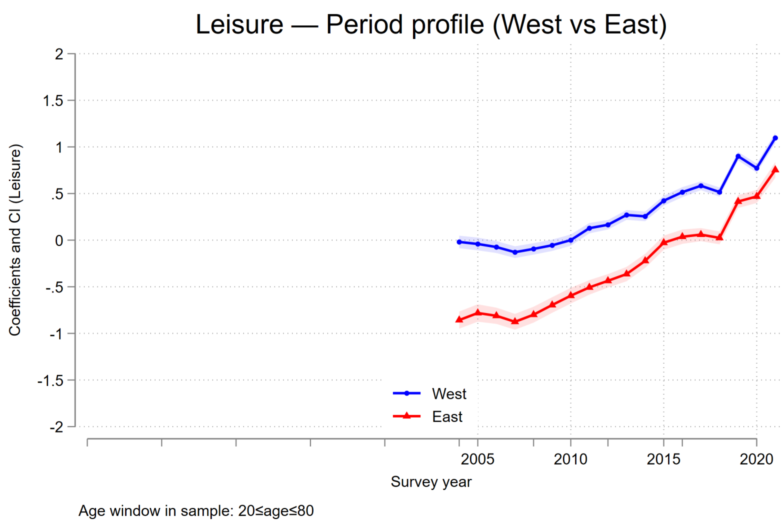


Figure 37: Satisfaction with Leisure (East vs West — Cohort Profile)

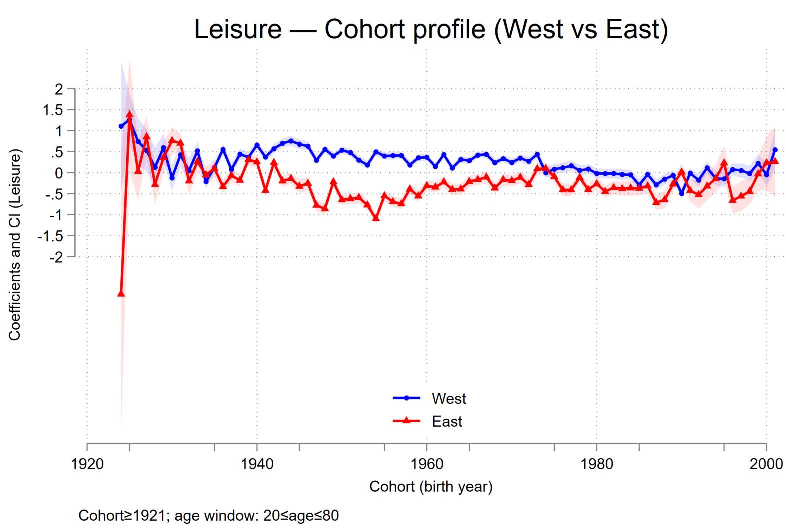


Figure 38: Satisfaction with Leisure (Migration Background — Age Profile)

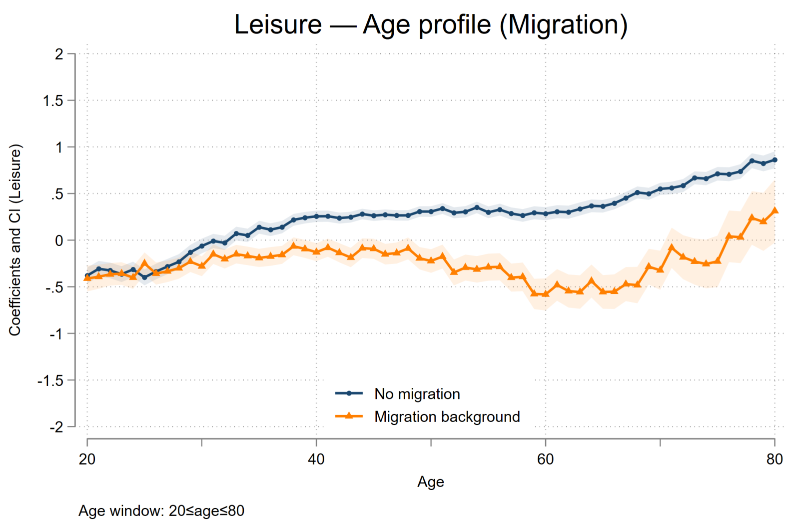


Figure 39: Satisfaction with Leisure (Migration Background — Period Profile)

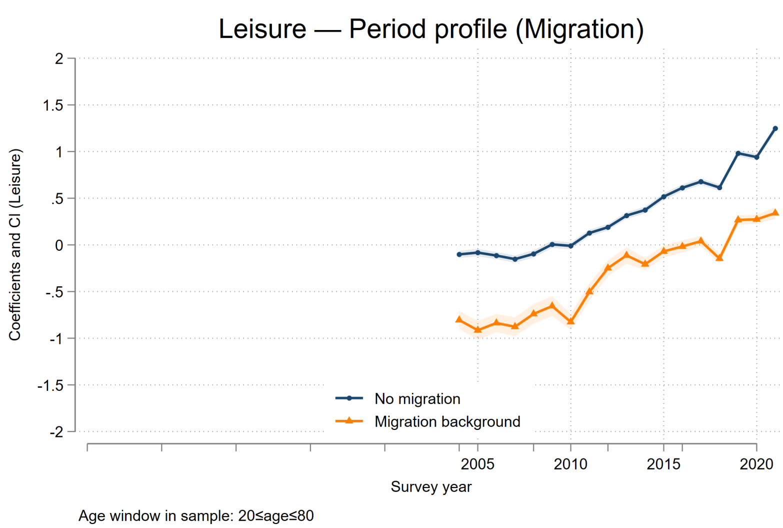
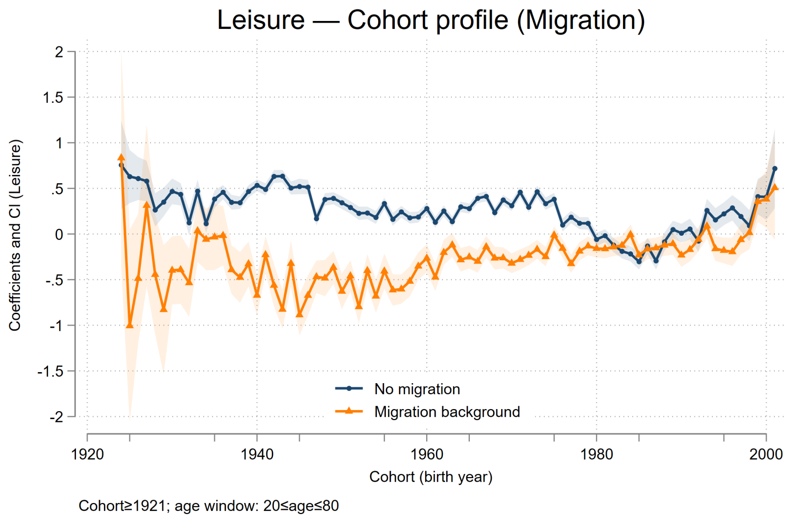


Figure 40: Satisfaction with Leisure (Migration Background — Cohort Profile)



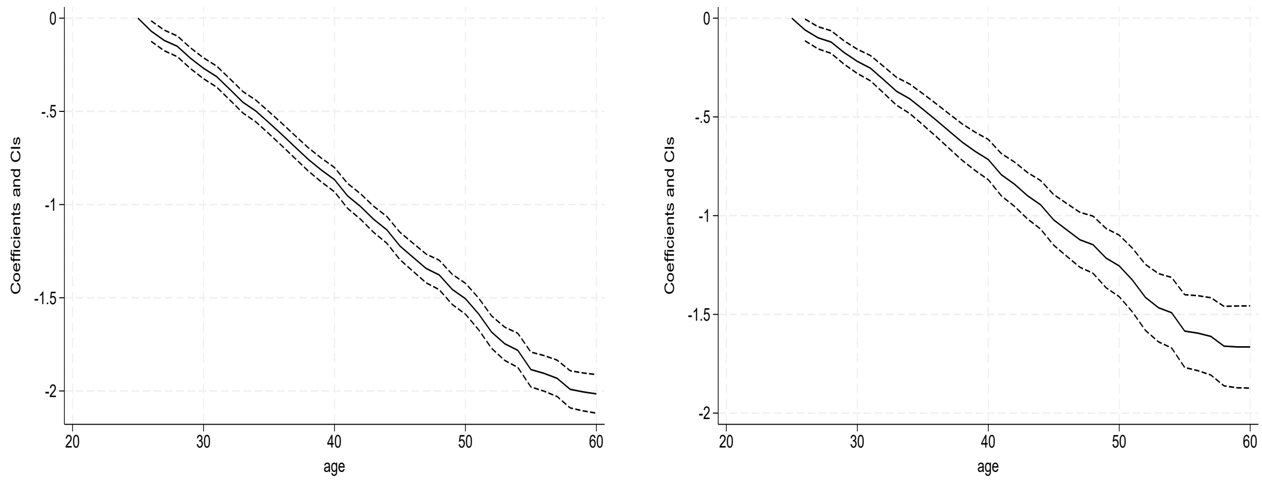
6.1 Robustness Check

Maximum Entropy (ME) Method versus Deaton and Parxon.
The DP approach (Deaton and Paxson, 1994) achieves identification by imposing linear restrictions on the period effects. Specifically, the estimated period component is constrained to be orthogonal to a linear trend and to sum to zero. Econometrically, this allocates the secular trend in the data to age and cohort dimensions, leaving period to capture only cyclical or transitory deviations. While straightforward and computationally convenient, this solution rests on strong functional-form assumptions about the structure of period variation.
By contrast, the ME approach (Browning, Crawford and Knoef, 2010) dispenses with these linear restrictions and instead invokes an information-theoretic criterion. When the dependent variable has finite support in the population, the estimator selects the age, period and cohort parameters that maximize the entropy of the implied distribution subject to the APC structure. Identification is thus achieved through an extremum principle rather than exclusion restrictions. This approach has the advantage of avoiding arbitrary trend-allocation rules but requires knowledge of the natural bounds

of the outcome variable and involves more intensive numerical optimization. In sum, the DP estimator relies on functional constraints to "purge" linear trends from the period component, whereas the ME estimator leverages the boundedness of the outcome and entropy maximization to obtain point identification. The choice between them reflects a trade-off between general applicability and the strength of identifying assumptions.
In practice, the graphical representation of estimated APC effects will often appear similar across the two methods when the outcome variable exhibits relatively smooth trajectories and the bounded nature of the dependent variable is not binding in the relevant range. In such cases, the ME estimator's entropy criterion will allocate variation in a manner that is close to the trend restrictions imposed by the DP approach, yielding nearly parallel age, period, and cohort profiles. Differences between the two methods are more likely to emerge when outcomes approach their natural bounds or when the period dimension carries a strong linear component. Under those conditions, the DP estimator, by construction, forces the linear trend into age and cohort, potentially understating systematic period effects, whereas the ME estimator can attribute more variation to period while remaining consistent with the bounded support of the dependent variable. Consequently, divergence in APC graphs across methods is most pronounced in settings with bounded outcomes that are highly skewed or in samples where secular period trends are empirically salient.

7 Appendix

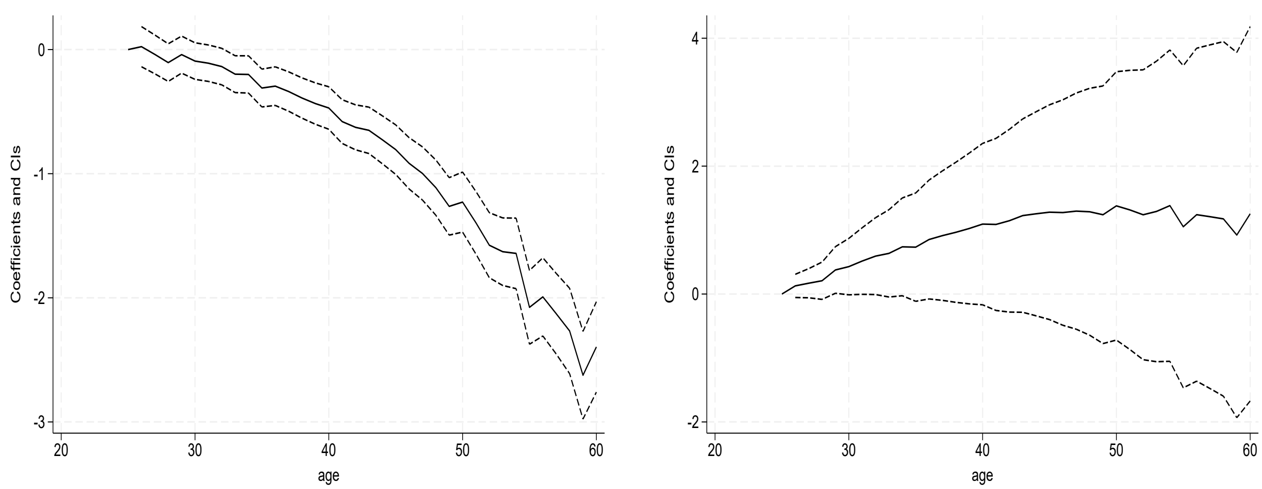
Figure 41: Age effects for *sat_work* (25–60)



(a) DP method

(b) ME method

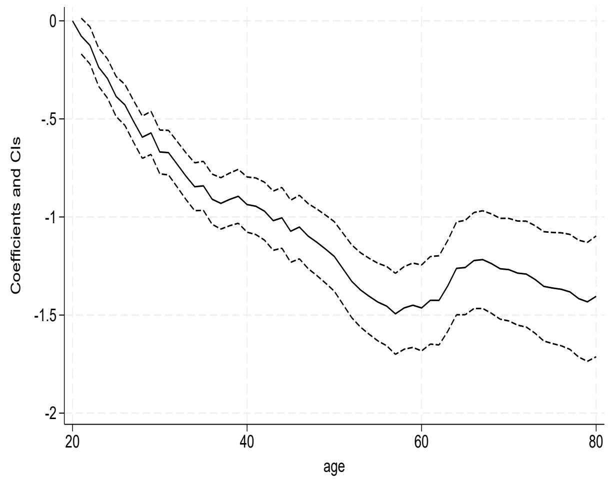
Figure 42: Age effects for *sat_commute* (25–60)



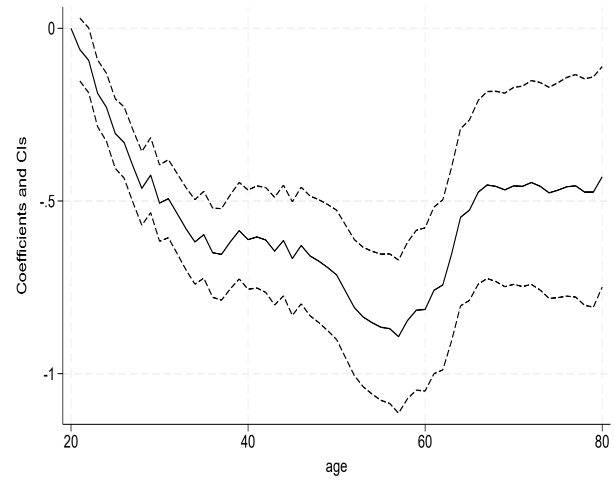
(a) DP method

(b) ME method

Figure 43: Age effects for *sat_health* (≤80)

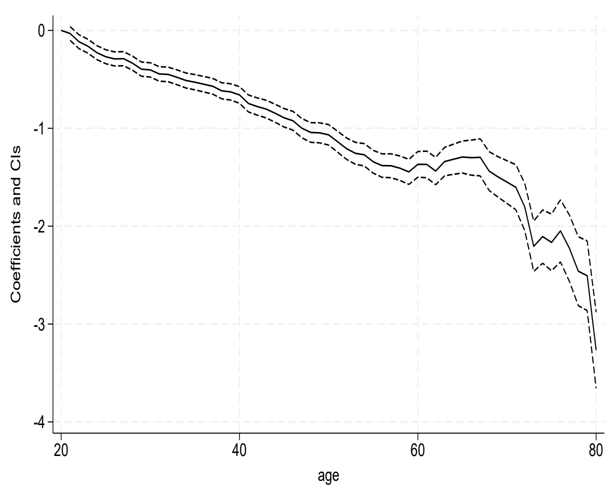


(a) DP method

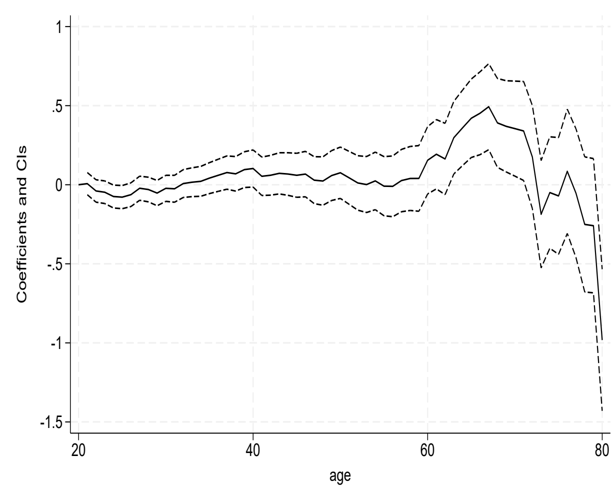


(b) ME method

Figure 44: Age effects for *sat_income* (≤80)

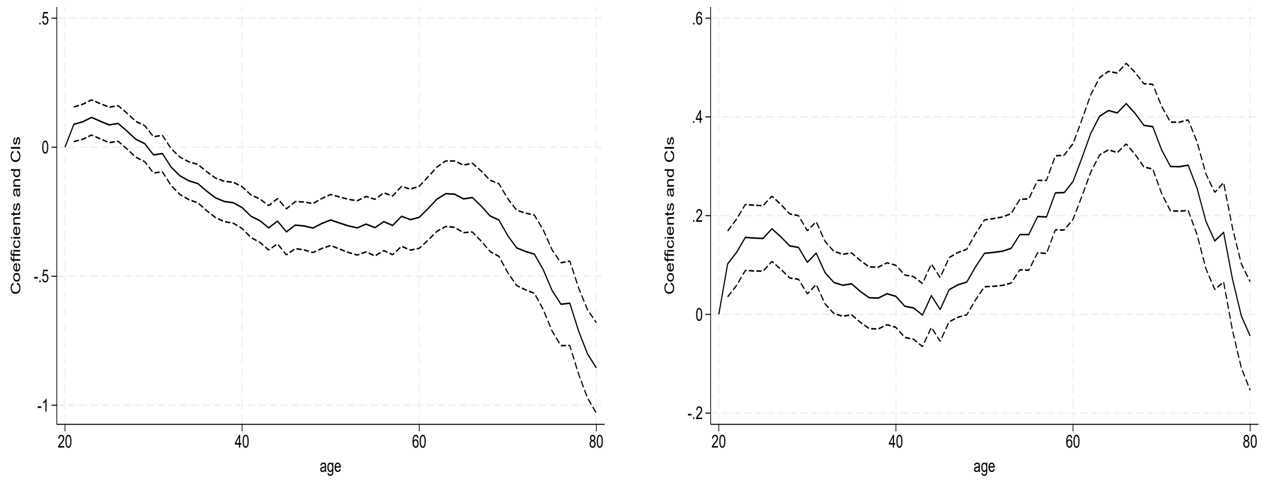


(a) DP method



(b) ME method

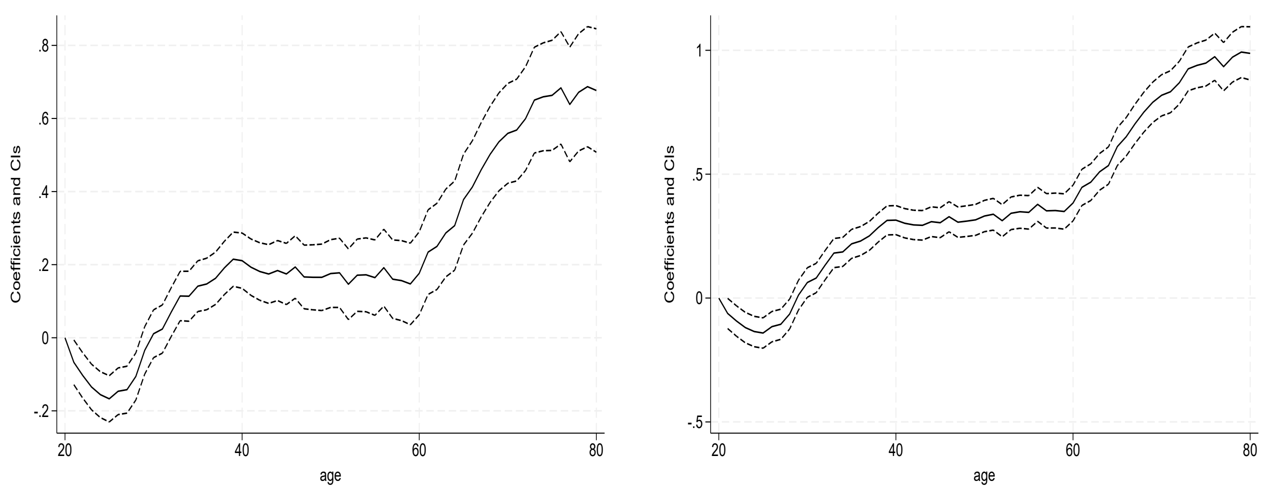
Figure 45: Age effects for *sat_housework* (≤80)



(a) DP method

(b) ME method

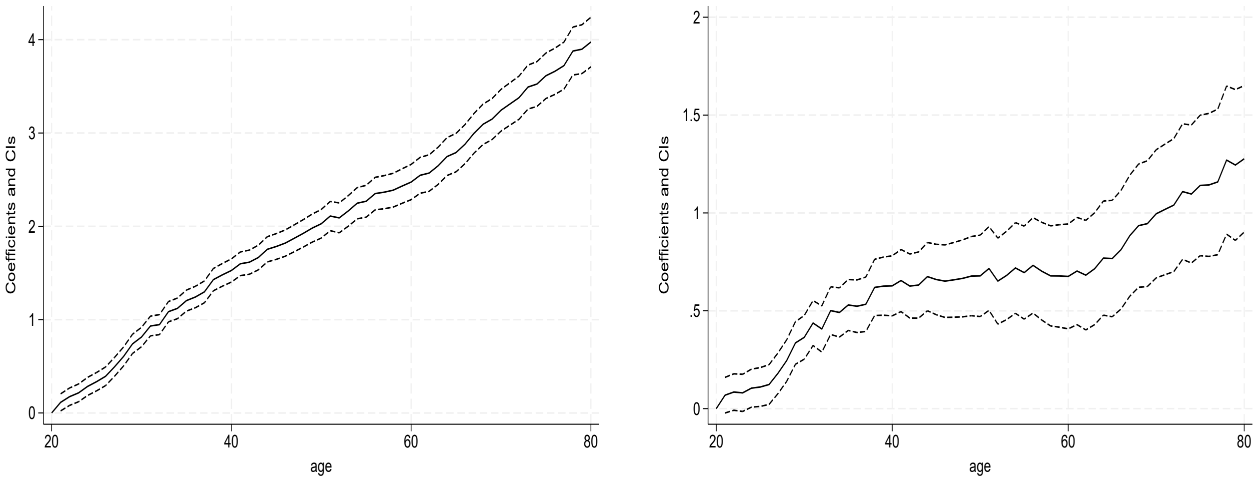
Figure 46: Age effects for *sat_childcare* (≤80)



(a) DP method

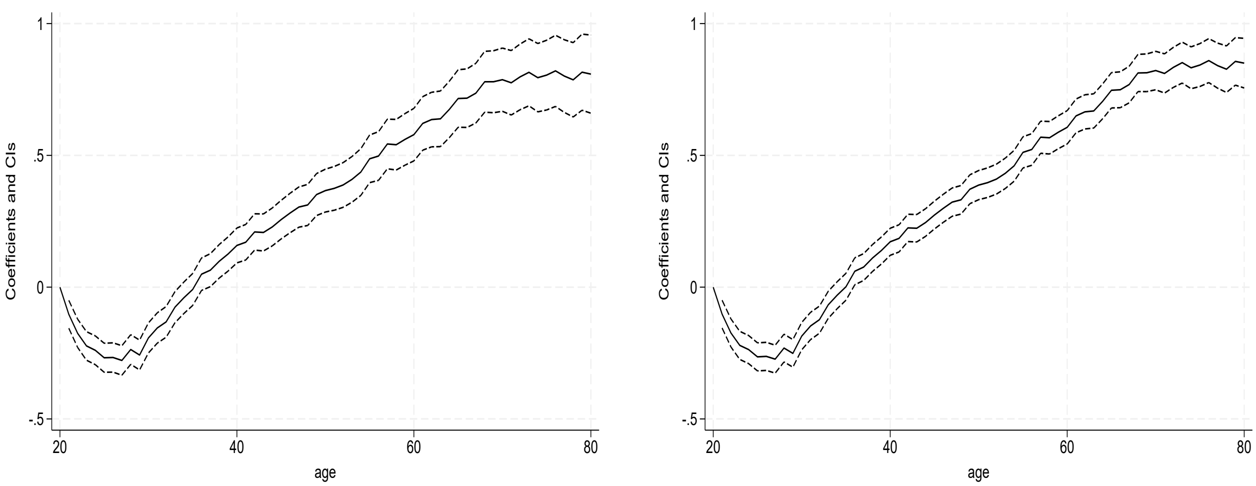
(b) ME method

Figure 47: Age effects for *sat_leisure* (≤80)



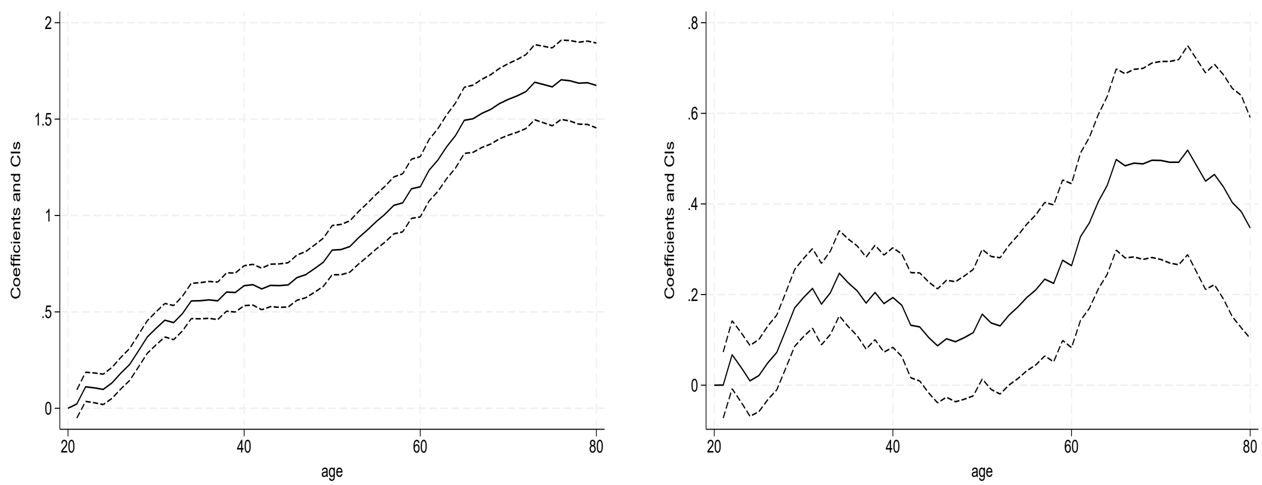
(a) DP method (b) ME method

Figure 48: Age effects for *sat_housing* (≤80)



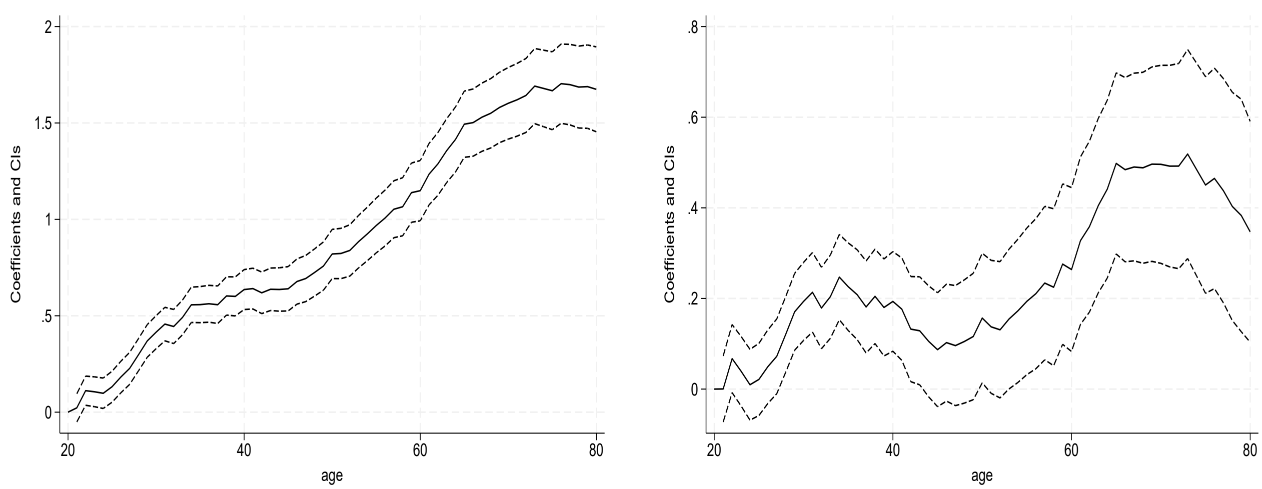
(a) DP method (b) ME method

Figure 49: Age effects for *sat_healthcare* (≤80)



(a) DP method (b) ME method

Figure 50: Age effects for *sat_enviornment* (≤80)



(a) DP method (b) ME method